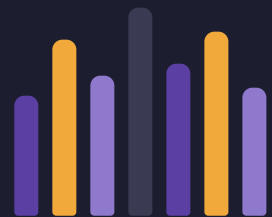




TESSERA POS · WINES & SPIRITS

# Demo Walkthrough & Test Report

How to prepare and run a client demo of the whole system, role by role, with the results of the end-to-end walkthrough carried out on 27 September 2026 and the issues found and fixed.



Version: main after the demo-walkthrough merge · Environment: local demo database (APP\_ENV=local)  
M-PESA and KRA eTIMS run on demo drivers: no real money moves and nothing is sent to KRA.

# 1. What this document covers

Tessera POS is now complete for the MVP and the approved Phase 2 items: catalogue, stock ledger, till and shifts, open bottles (tots), M-PESA, purchasing, eTIMS, reports, the offline till, customers, settings, sign-in security, dashboard KPIs, credit and supplier accounts, notifications, promotions, and expenses and licences. This guide:

- prepares a demo database with realistic data (section 2);
- lists the demo logins and till PINs (section 3);
- gives a step-by-step demo script for each role (section 4);
- records what the walkthrough tested and what happened (section 5);
- lists the issues found, with their fixes (section 6), and the known limits (section 7).

**Integrations are in demo mode.** Real Safaricom Daraja and KRA eTIMS will be connected once the client provides credentials and the VSCU/OSCU specification. Until then, the demo M-PESA driver confirms payments with a made-up receipt number, and the demo eTIMS driver signs invoices with made-up control codes. Every screen behaves as it will in production.

## 2. Preparing the demo

### 2.1 Start the system

```
# API (port 8010) and scheduler, from backend/  
php artisan serve --host=127.0.0.1  
php artisan schedule:work  
  
# Web app (port 3010), from frontend/  
pnpm dev
```

Open **http://localhost:3010**. Use `localhost`, not `127.0.0.1`, so the sign-in cookie is shared with the API. The scheduler sends the eTIMS outbox, alerts and the daily summary. Keep it running during a demo.

### 2.2 Load the demo showcase

```
php artisan db:seed --class=DemoShowcaseSeeder --force
```

This works only with `APP_ENV=local` and is safe to run again before every demo. It does the following:

| Item            | What the seeder does                                                                                                                                             |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Two-step login  | Switched off for the owner so the demo needs no phone. Tessera Support always keeps it. Turn it back on under Settings → Staff and roles.                        |
| Till PINs       | Resets the demo PINs to those in section 3, in case they were changed while testing.                                                                             |
| Customer credit | Switches customer credit on (any cashier may sell on account within the limit). <i>Demo Lounge &amp; Grill</i> gets a KES 30,000 limit and 30-day terms.         |
| Promotions      | <i>Wine weekend: buy 2, 10% off</i> (any wine, 2 or more bottles).<br><i>Jameson happy hour</i> (KES 300 off Jameson 750 ml, 17:00–20:00). Both are approved.    |
| Licences        | The Nairobi County liquor licence expires in 20 days: it triggers an alert and is printed on the receipt. The single business permit is valid for 9 more months. |

Everything else (purchase orders, goods received, sales, payments, expenses) is done in the app during the demo, so the audience sees every document being created.

### 3. Demo accounts

| Person          | Role           | Back-office sign-in                              | Till PIN |
|-----------------|----------------|--------------------------------------------------|----------|
| Business Owner  | Owner          | owner@tessera.test / password                    | 1470     |
| Wanjiru Mwangi  | Branch Manager | manager@tessera.test / password                  | 4826     |
| Otieno Kamau    | Cashier        | till only                                        | 2580     |
| Amina Hassan    | Cashier        | till only                                        | 3691     |
| Njeri Wambui    | Storekeeper    | njeri@tessera.test / password                    | —        |
| Faith Achieng   | Accountant     | accountant@tessera.test / password               | —        |
| Tessera Support | Tessera Admin  | support@tessera.test / password (two-step login) | —        |

These accounts exist only in local and testing databases. They are never created in production.

#### What each role sees

| Role           | Menu                                                                                                                                                                                                                                                     |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Owner          | Everything, including Branches, Users & roles, Settings, Messages sent and Audit log.                                                                                                                                                                    |
| Branch Manager | Dashboard, POS till, sales and shifts, catalogue including promotions, all inventory, purchasing, supplier and customer accounts, M-PESA reconciliation, expenses, eTIMS, licences, reports, settings (branch level), audit log. No user administration. |
| Accountant     | Dashboard, sales, products (view), stock views, purchasing, supplier and customer accounts, M-PESA reconciliation, expenses, eTIMS, licences, reports, audit log. No till, promotions, settings or users.                                                |
| Storekeeper    | Products, stock on hand, adjustments, transfers, counts, open bottles, stock ledger, purchase orders, supplier invoices and returns, suppliers. The dashboard shows stock tiles only.                                                                    |
| Cashier        | The till only (PIN sign-in on a paired till device).                                                                                                                                                                                                     |

### 4. Demo script

The script below takes about 30–40 minutes. It follows a bottle from the supplier to the customer and then to the books. Each step names who is signed in.

#### 4.1 Owner: the business at a glance (5 min)

1. Sign in as the owner. The dashboard greets them and shows today's KPI tiles: sales, cash, M-PESA/card, gross profit, items on sale, low stock, stock value at cost, shrinkage, amounts owed by customers and to suppliers, and eTIMS invoices not yet signed.
2. Point out **Needs attention** (low stock, the licence expiring in 20 days, exceptions) and **Shifts open now**.
3. Open the bell: the liquor-licence and low-stock alerts are there.
4. Settings: show the business, receipt, till and staff sections, and how a change can be undone from its history.

#### 4.2 Purchasing: order, receive, invoice, pay (8 min)

1. **Storekeeper or owner:** raise a purchase order for the low-stock item from the suggested supplier.

2. **Owner:** try to approve it. It is refused with "Someone else must approve this — you raised it." This is the maker-checker rule.
3. **Manager:** approve the order.
4. **Storekeeper:** receive it, with received and damaged quantities per line. A goods-received note (e.g. `MAIN-GRN-000001`) is posted and stock rises in the stock ledger.
5. **Accountant:** record the supplier invoice against the delivery. Tessera matches it to the order and the receipt.
6. **Accountant or manager:** record a supplier payment. Supplier accounts show the new balance and the payables aging.

### 4.3 The till: a cashier's shift (12 min)

1. Open <http://localhost:3010/till> on the paired till. Choose *Otieno* and enter PIN **2580**, then start the shift with the opening float.
2. The first screen shows best sellers. Add two bottles of wine: the *Wine weekend* promotion takes 10% off automatically. For example,  $2 \times \text{KES } 1,050 - \text{KES } 210 = \text{KES } 1,890$ .
3. Spirits ask the cashier to confirm the customer's age.
4. Pay by **M-PESA**: the demo prompt is confirmed with a receipt number. The receipt shows the promotion, the liquor-licence line and the (demo) eTIMS signature and QR details.
5. Choose *Demo Lounge & Grill* as the customer and pay **On account**. The sale goes on their credit account within the KES 30,000 limit.
6. **Payout:** pay KES 500 from the drawer for a small expense. A manager witnesses it with their PIN.
7. **Lock** the till. A wrong PIN is refused, the cart is kept, and Otieno's PIN unlocks it. The till also locks itself after 2 minutes idle.
8. **Return:** refund one item from an earlier sale with a reason. A return note (e.g. `MAIN-RET-000002`) is created and the stock goes back.
9. **Close the shift:** count the drawer. Tessera shows the expected cash (float + cash sales – refunds – payouts) against the counted amount and records any variance for the manager's sign-off.

### 4.4 Accounts and expenses (6 min)

1. **Accountant:** in *Customer accounts*, open Demo Lounge & Grill, record a payment and show the statement and aging (payments settle the oldest sales first).
2. **Accountant:** record an expense (e.g. electricity) from petty cash or the bank. It waits for approval.
3. **Owner:** an owner cannot approve an expense they raised themselves. The **manager** approves it and it appears in the expenses report.
4. **Licences & permits:** show the expiry dates, the 20-day warning and the licence line on receipts.

### 4.5 Reports and control (5 min)

1. Reports: sales summary, cash-ups, expenses, receivables and payables aging, stock valuation and shrinkage. Each can be printed or exported to CSV.
2. eTIMS monitor: every sale and return with its signing status. A rejected document raises an alert.
3. Audit log: every approval, price change, payout, lock and setting change, with who did it and when.
4. Messages sent: the daily summary and alerts that would go by SMS or WhatsApp (logged in demo mode).

## 5. Walkthrough results (27 September 2026)

The full script above was run on the local demo database through the real web app and API, signed in as each role. Results:

| Area | What was done | Result |
|------|---------------|--------|
|------|---------------|--------|

|                     |                                                                                                                                                                                                                  |      |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Sign-in             | Owner signs in without a code after the showcase seeder. Every role's menu matches its permissions. Requests outside the role are refused with HTTP 403.                                                         | PASS |
| Maker-checker       | The owner could not approve their own purchase order or expense; the refusal now gives the reason. The manager approved both.                                                                                    | PASS |
| Purchasing          | Order approved → received as MAIN-GRN-000001 → supplier invoice RVSD-88412 matched → supplier payment SP-000001 . The supplier balance fell to KES 92,819.20.                                                    | PASS |
| Till: selling       | Best sellers shown. Wine promotion $2 \times 1,050 - 210 = 1,890$ . Age check shown. Demo M-PESA confirmed (receipt FKF6CY3DC7). The receipt shows the promotion, the licence line and the demo eTIMS signature. | PASS |
| Till: on account    | Sale MAIN-S-000010 of KES 3,100 went on Demo Lounge & Grill's account within the limit.                                                                                                                          | PASS |
| Till: payout, lock  | KES 500 payout witnessed by the manager. After locking, the API refused requests (HTTP 423) and a wrong PIN was refused. After unlocking, nothing leaked into product search.                                    | PASS |
| Till: return, close | Return MAIN-RET-000002 refunded KES 945. At shift close, expected cash KES 3,555 equalled counted cash, with payouts of KES 500.                                                                                 | PASS |
| Customer account    | Payment MAIN-RCP-000001 of KES 1,000. Balance KES 2,100, available KES 27,900. Statement: sale 3,100, payment -1,000. Aging correct.                                                                             | PASS |
| Expenses            | MAIN-EXP-000001 approved by the manager. MAIN-EXP-000002 left pending to show the queue.                                                                                                                         | PASS |
| Dashboard           | Owner, accountant and storekeeper dashboards load with the tiles for their role. Receivables, payables, expenses, licences and exceptions agree with the screens behind them.                                    | PASS |
| Notifications       | The bell showed 2 unread alerts: licence expiry and low stock.                                                                                                                                                   | PASS |
| Reports             | Sales summary, cash-ups, expenses, receivables aging and payables aging all loaded. Promotions list showed KES 210 discount given.                                                                               | PASS |
| Automated checks    | After the fixes: backend suite (188 tests, 1,517 assertions, plus new checks for refusal reasons and malformed ids), Pint code style, ESLint and the UI-conventions check, TypeScript, and the production build. | PASS |

## 6. Findings and fixes

| # | Finding                                                                                                                                                             | Fix                                                                                                 | Status |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|
| 1 | Refusals always read "You are not allowed to perform this action.", even when there was a specific reason such as "Someone else must approve this — you raised it." | The API error handler now keeps specific reasons and replaces only the framework's generic message. | FIXED  |
| 2 | A malformed record id in a URL (e.g. /suppliers/undefined/account ) gave a server error.                                                                            | It now returns "Resource not found." (404).                                                         | FIXED  |
| 3 | Dashboard "Shifts open now" showed only a time ("since 5:35 am") for a shift opened the day before.                                                                 | The date is shown when the shift did not open today: "since 26 Sep, 5:35 am".                       | FIXED  |
| 4 | After a sale fully on account, the till said "Paid in full".                                                                                                        | It now says the sale went on the customer's account.                                                | FIXED  |

|    |                                                                                           |                                                                                                                                 |              |
|----|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------|
| 5  | The till reloaded the favourites (best sellers) several times when the screen opened.     | They now load once; the offline copy is used only when the till is offline.                                                     | <b>FIXED</b> |
| 6  | There was no demo accountant login.                                                       | Faith Achieng ( <code>accountant@tessera.test</code> ) was added to the demo accounts.                                          | <b>FIXED</b> |
| 7  | Demo till PINs had drifted during earlier testing.                                        | The showcase seeder resets them before each demo.                                                                               | <b>FIXED</b> |
| 8  | The demo owner is named "Business Owner", so the greeting reads "Good morning, Business". | Demo data only. Rename the owner under Users & roles before a client demo if wanted.                                            | <b>NOTE</b>  |
| 9  | Today's "Cash" tile can be negative when a refund of an M-PESA sale is paid in cash.      | By design: refunds are paid from the drawer, and the tile shows the drawer's net cash. M-PESA refunds (B2C) are a Phase 2 item. | <b>NOTE</b>  |
| 10 | An older shift from earlier testing (Amina, Till 1) is still open in the demo database.   | Close it from the till before a client demo, or leave it to show the "open since yesterday" warning.                            | <b>NOTE</b>  |

## 7. Known limits and next steps

### Waiting on the client

- Real Safaricom Daraja (STK push, C2B confirmations): needs the shortcode, passkey and API keys.
- Real KRA eTIMS (VSCU/OSCU): needs the device registration, specification and sandbox access.
- SMS/WhatsApp provider for alerts (currently logged in demo mode).

### Needs a decision first

- Loyalty points: consent rules and alcohol-promotion rules need a legal check.
- Other Phase 2 / future items: licensed trading-hour rules, M-PESA B2C refunds, debit notes, card terminals, expiry/batch tracking, excise stamps.
- Automated end-to-end tests for the till (browser tests).

**Before going live:** switch two-step login back on for owners, remove the demo accounts (production seeding never creates them), set `MPESA_DRIVER` and `ETIMS_DRIVER` to the real drivers, and run the scheduler as a service.